Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/01/2024
PI Name:	Wu, Joseph M.D., Ph.D.		
Project Title:	Assessing Long-Term Effects of Radiation Exposure in Engineered Heart & Vascular Tissues		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	joewu@stanford.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	650-736-2246
Organization Name:	Stanford University		
PI Address 1:	265 Campus Drive, G1120B		
PI Address 2:			
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305	**Congressional District:**	16
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	09/27/2022	**End Date:**	09/26/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	April 2024 update regarding changes to the list of Co-Investigators: Dr. Peter Grabham, Dr. Vandana Verma, and Dr. Wing Hung Wong are no longer included on the list of Co-Investigators. Per the PI Team: At the beginning of grant activation, it was realized that phase 1 (optimizing 3D tissue culture condition) would require exclusive efforts from the Stanford side; thus, key personnel have been adjusted based on the project's requirements. Dr. Wong recently retired. The PI team will have internal collaborators (e.g., Dr. Michael Snyder) to support statistical analysis of the function/sequencing data sets to be generated in Phase 2. (Ed., 4/24/24)		
COI Name (Institution):	Beheshti, Afshin Ph.D. (NASA Ames Research Center) Moding, Everett M.D., Ph.D. (Stanford University)		

Grant/Contract No.:	80ARC022CA003
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Deep space is full of myriad risks and exposure to ionizing radiation is one of the biggest health risks associated with spaceflight. The conventional risk assessment in animal models or immortalized cell lines does not fully recapitulate changes in human heart function upon radiation exposure. To examine the individual susceptibility to radiation-induced cardiovascular disease and an accurate assessment of tissue function, we propose to use human induced pluripotent stem cells (iPSCs) and generate 3D heart-like and vessel like tissues from genetically diverse human subjects. The Stanford iPSC biobank has established the largest academic center biorepositories with >1,500 human iPSCs from various demographics. These already established iPSC lines have been extensively characterized at both genotype and phenotype levels. We will take 8 genetically diverse iPSC lines with an equal distribution in sex and ethnicity (Caucasians, African Americans, Hispanics, and Asians; n=2/each) and differentiate them into diverse cardiovascular cell types. Using our established robust protocols, we will assemble 3D engineered heart tissues (EHTs) and engineered vascular tissues (EVTs). EHTs are composed of iPSC-cardiomyocytes, iPSC-endothelial cells, and iPSC-cardiac fibroblasts, whereas EVTs are composed of iPSC-endothelial cells, iPSC-macrophages, and iPSC-smooth muscle cells. We will expose 3D EHTs and EVTs with X-ray radiation at total doses of 0, 1, 2, and 5 Gy delivered at single fraction. We will analyze the changes in tissue functions (e.g., contractility – EHTs, vessel integrity – EVTs) over 6 months post irradiation. We will collect culture medium for biomarker analysis (e.g., cytokines, chemokines, exosome-derived microRNAs). At endpoint, we will analyze the transcriptomic and epigenomic profiles at single cell levels with scRNA-seq and scATAC-seq. Understanding dose and dose-rate effective factor values for tissue specific endpoints across different tissue types to radiation exposure has been an interest to NASA and the space biology research community. The implementation of a 3D iPSC-based tissue platform to monitor long-term effects of radiation exposure will facilitate the deployment of personalized countermeasures for astronauts or future space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Outcomes: Our robust and well-validated 3D cardiovascular tissue models with novel cell tracing approaches will enable long-term (>6 months) and non-invasive monitoring of (i) tissue functions and (ii) cell composition after radiation exposure. Deliverables: Establishing the iPSC-based platform can be extended by performing personalized assessment for relevant populations (e.g., astronauts) and in different organs (e.g., brain, liver). Generating multi-omics data from a larger cohort will improve our understanding of genetic determinants of complex clinical traits and responses.
Task Progress:	[Ed. Note: For complete citations, see References.] 1. Finalization in induced pluripotent stem cells (iPSC) selection for the study: In the interest of achieving robust statistical significance for assessing potential effects related to ethnicity and sex, the initial plan of utilizing a diverse iPSC panel has been refined. We are using 4 iPSC lines from 2 males and 2 females with 2 ethnicities (Caucasian, African American). We have seeded them in the culture plates for serial passage. iPSCs at the early passage (<passage 10) have the viral genes from reprogramming (CytoTune™-iPS 2.0 Sendai Reprogramming Kit, #A16517, Thermo Fisher Scientific Inc.) and the epigenetic signatures from the original tissues or cell types. Serial passage up to 16 passages significantly reduce the expression of Sendai viral genes and the original tissue epigenetic signatures (PMID: 20644536). 2. Generation of iPSC-derivatives: The initial mesoderm differentiation process is uniform for all iPSC-derivatives for this study. The successful generation of high-quality cardiovascular cell types hinges on the efficacy of mesoderm commitment during the first 2 days of differentiation. To achieve this, we've implemented a small molecule-based protocol (PMID: 22645348). Specifically, we activate the canonical Wnt signaling pathway using CHIR99021 (a GSKß inhibitor) at concentrations typically ranging from 6 to 8 µM. After careful optimization, we have finalized the optimization of the mesoderm induction conditions for 8/8 lines and have stocked all cell types (cardiomyocytes – CMs, endothelial cells – ECs, cardiac fibroblasts – CFs, smooth muscle cells – SMCs) that will be used in generating engineered heart tissues (EHTs) and engineered vascular tissues (EVTs) from 8 lines. 3. Generation of EHTs & EVTs: We have established workflow to generate EHTs made up of iPSC-CMs, CFs, and ECs at defined ratios, and numerous studies have been conducted using 3D EHTs. For EVTs, we will take commercially available microfluidic platforms to generate EVTs, and we have explored two systems: iBIDI and Emulate. Both systems allow researchers to generate blood flow driven shear stress (to mimic the physiological state), yet Emulate offers the double chamber systems to study cellular crosstalk. 4. Optimization of 3D culture medium: We use Roswell Park Memorial Institute (RPMI) 1640 with serum free B27™ Supplement, which we routinely use to culture iPSC-CMs. We previously demonstrated that the long-term culture of iPSC-CMs in RPMI1640 + B27 (>6 months) improved the energy production and contractile function over time. However, we observed the population of ECs regress over time when EHTs are cultured in RPMI1640 + B27. Three months of culturing EHTs in RPMI1640 + B27 decreased the ECs' composition from 15% to <5%. As the vascular cells play an important role in normal heart function, and multiple literatures stated an importance of vascular cells in mediating radiation induced cardiac dysfunction, we will optimize the culture medium to maintain both CMs and ECs. Our reporter construct (TNNTGFP-CDH5RFP-) enables us to trace the composition of ECs without invasive sampling. We will titrate the angiogenic growth factors (e.g., VEGF, FGF2) mix with EC growth medium (EGM2, Lonza) to titrate the cell composition of CMs and ECs. We have successfully identified the co-culture media (CCM) in chemically defined (serum-free) formulations to ensure reproducibility, robustness, and scalability. This report details our use of the reporter cell line iPSCTNNT2-eGFP; CDH5-mOrange to monitor the abundance and signal intensity of Cardiomyocytes (TNNT2) and Endothelial cells (CDH5) in response to various CCM formulations. Specifically, we focused on CCM compositions based on RPMI + B27, which serves as the baseline medium for Cardiomyocytes, in combination with selected components from endothelial growth media (EGM) (1 – hydrocortisone; 2 – IGF-1; 3 – Ascorbic Acid; 4 – EGF; 5 – Heparin; 6 – VEGF). Over the course of 30 days, we co-incubated Cardiomyocytes (CMs) with Endothelial cells (ECs),

changing the media every four days. We successfully identified eight CCM combinations that resulted in a significant population of functional cardiomyocytes with contractile capabilities. Notably, each of these successful formulations included ascorbic acid as a critical component for maintaining the long-term viability and functionality of cardiomyocytes.

REFERENCES

1. Polo JM, Liu S, Figueroa ME, Kulalert W, Eminli S, Tan KY, Apostolou E, Stadtfeld M, Li Y, Shioda T, Natesan S, Wagers AJ, Melnick A, Evans T, Hochedlinger K. Cell type of origin influences the molecular and functional properties of mouse induced pluripotent stem cells. Nat Biotechnol. 2010 Aug;28(8):848-55. https:// ; PMID: 20644536; PMCID: PMC3148605

2. Lian X, Hsiao C, Wilson G, Zhu K, Hazeltine LB, Azarin SM, Raval KK, Zhang J, Kamp TJ, Palecek SP. Robust cardiomyocyte differentiation from human pluripotent stem cells via temporal modulation of canonical Wnt signaling. Proc Natl Acad Sci U S A. 2012 Jul 3;109(27):E1848-57. https:// ; PMID: 22645348; PMCID: PMC3390875

Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Cao X, Thomas D, Whitcomb LA, Wang M, Chatterjee A, Chicco AJ, Weil MM, Wu JC. "Modeling ionizing radiation-induced cardiovascular dysfunction with human iPSC-derived engineered heart tissues." J Mol Cell Cardiol. 2024 Mar;188:105-7. https://doi.org/10.1016/j.yjmcc.2023.11.012 ; PMID: 38431383; PMCID: PMC10961094 , Mar-2024
Articles in Peer-reviewed Journals	Caudal A, Snyder MP, Wu JC. "Harnessing human genetics and stem cells for precision cardiovascular medicine." Cell Genom. 2024 Feb 14;4(2):100445. https://doi.org/10.1016/j.xgen.2023.100445 ; PMID: 38359791; PMCID: PMC10879032 , Feb-2024
Articles in Peer-reviewed Journals	Whitcomb LA, Cao X, Thomas D, Wiese C, Pessin AS, Zhang R, Wu JC, Weil MM, Chicco AJ. "Mitochondrial reactive oxygen species impact human fibroblast responses to protracted gamma-ray exposures." Int J Radiat Biol. 2024 Apr 17;1-13. https://doi.org/10.1080/09553002.2024.2338518 ; PMID: 38631047 , Apr-2024